

Project Proposal

Mastering Connect 4 via Deep Reinforcement Learning and Monte Carlo Tree Search

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1. Introduction

This project proposes the development of a software agent to master Connect 4, a game with a state complexity of 4.5×10^{12} . The objective is to implement a Deep Reinforcement Learning (DRL) system inspired by AlphaZero. By combining Monte Carlo Tree Search (MCTS) with a Convolutional Neural Network (CNN), the agent will learn optimal strategies through self-play, approximating the Nash Equilibrium without human-derived heuristics.

2. Game Theoretic Framework

Connect 4 is formally modeled as a finite, deterministic, zero-sum game with perfect information.

- **Definition:** $G = (N, S, A, T, u)$, where players $N = \{1, 2\}$ compete for utility $u \in \{+1, -1, 0\}$.
- **Solution Concept:** As a solved game, a pure strategy Subgame Perfect Nash Equilibrium exists (proven as a First-Player Win).
- **Goal:** The agent seeks to approximate the optimal value function $V^*(s)$ via a function approximator $f_{\theta(s)}$, rather than exhaustive minimax search.

3. Methodology

The implementation will replicate the core AlphaZero loop using Python and PyTorch:

3.1. Neural Network (f_{θ})

A Residual CNN takes the board state s as input and outputs:

- **Policy Head (p):** A probability distribution over valid moves.
- **Value Head (v):** A scalar estimate $[-1, 1]$ of the current state's win probability.

3.2. MCTS Guidance

Instead of traditional Minimax, MCTS is used for lookahead. The search balances exploration and exploitation using the PUCT algorithm, guided by the network's policy p .

3.3. Self-Play Training

The agent plays games against itself. The MCTS search probabilities π and final game outcome z are used to update network weights θ by minimizing the loss:

$$L = (z - v)^2 - \pi^{\top} \log p + c \|\theta\|^2$$

4. Evaluation Metrics

The agent's performance will be evaluated against three benchmarks:

1. **Baselines:** Win rate vs. Random and Greedy (1-step lookahead) agents.
2. **Traditional AI:** Performance against a standard Minimax agent with Alpha-Beta pruning (fixed depth).
3. **Learning Stability:** Analysis of loss convergence and Elo rating progression during the training phase.

5. Proposed Final Deliverable Format: Written Report

Given the algorithmic complexity and mathematical density of this project, I respectfully request to submit a comprehensive written report in lieu of a live presentation.